Silicon Samples Worksheet

Sean Murphy

9 October, 2026

One persona, many draws

Each model was asked the same question twenty times, using `draw` values 0 to 19 (`01-draws.py`), once at `temperature=1.0` and once at `temperature=0`. The persona was "You are a woman in your thirties who lives in the Midwest of the United States and finished high school." The question asked whether she would vote for the Democratic or Republican candidate, someone else, or not vote in the House election if it were held today, and how confident she was from 0 to 100.

The models were:

- DeepSeek: `deepseek/deepseek-v4-flash-0731`, through OpenRouter
- GLM: `z-ai/glm-5.3-flash`, through OpenRouter
- Muse: `meta/muse-glimmer-30b`, through OpenRouter
- Gemma: `gemma4:12b`, run locally with Ollama. This is an extra model for comparison and is not part of the worksheet's model set.

Table 1: Votes from twenty draws per model and temperature, one persona.

Model	Temperature	Democratic	Republican	Other	Would not vote
DeepSeek	1	18	2	0	0
	0	16	4	0	0
GLM	1	4	16	0	0
	0	1	19	0	0
Muse	1	7	0	0	13
	0	0	0	0	20
Gemma	1	6	0	5	9
	0	0	0	1	19

Table 2: Share of the twenty draws that gave the modal answer, by temperature (T). The modal answer is the same at both temperatures for every model.

Model	Modal answer	Modal share (T = 1)	Modal share (T = 0)
DeepSeek	Democratic	90%	80%
GLM	Republican	80%	95%
Muse	Would not vote	65%	100%
Gemma	Would not vote	45%	95%

Observations:

- The same persona gets a different modal answer depending on the model: Democratic for DeepSeek, Republican for GLM, and "Would not vote" for Muse and Gemma. Each model's modal answer is the same at both temperatures.
- At `temperature=1.0`, how concentrated the answers are varies by model, from 90% for DeepSeek to 45% for Gemma. Gemma is the only model that answered "Other".
- `temperature=0` did not make the answers identical. Only Muse gave the same answer all twenty times. GLM and Gemma each gave one answer that differed from the rest, and DeepSeek gave four. For the three hosted models, a likely reason is that hosted inference is not exactly reproducible: requests are batched on GPUs, and OpenRouter can route each call to a different provider. Gemma ran on one machine with no routing and still varied, so `temperature=0` alone does not guarantee identical answers.
- DeepSeek's modal share was lower at `temperature=0` (80%) than at `temperature=1.0` (90%). With twenty draws, the standard error of a share near 85% is about eight points, so this difference is within noise.
- No draw at either temperature returned a null answer.